Mostafa A. Atalla

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RESEARCH INTERESTS

Medical and surgical robotics; soft and continuum robots; mechanically intelligent design and physical interaction; friction modulation, tribology, and safe, functional devices for minimally invasive interventions.

ACADEMIC APPOINTMENTS

2024–2026 **Postdoctoral Researcher** University of Twente, The Netherlands
Robotics and Mechatronics (RaM) Group, EEMCS. Research on mechanically intelligent soft robots for safe interactions in colonoscopy. Advisors: Prof. dr. ir. Gijs Krijnen.

2020–2026 **PhD Researcher** TU Delft, The Netherlands
Faculty of Mechanical Engineering. Dissertation: *'When Friction Meets Function: Engineering Medical Instrument with Friction to Glide, Grip and Propel'*. Promotors: Dr. M. Wiertlewski, Dr. ir. A. Sakes, and Prof. dr. ir. P. Breedveld.

EDUCATION

2018–2020 **MSc in Robotics Engineering** Worcester Polytechnic Institute, MA, U.S.A.
Fulbright Scholar. Advisor: Dr. Loris Fichera. Specialization: Medical Robotics. GPA: 4.0/4.0.

2009–2014 **BSc in Mechanical Design and Production Engineering** Cairo University, Egypt
Advisors: Prof. dr. Yasser H. Anis and Prof. dr. Sayed M. Metwalli. GPA: 3.3/4.0. Ranked 3rd out of 37 students.

PROFESSIONAL EXPERIENCE

2020 **Intern – Mechanical Design for Robotics and Navigation** Philips Research NA, Cambridge, MA, U.S.A.
Image Guided Therapy Department, March 2020–September 2020.

2019 **Research Intern – Mechanical Design** Philips Research NA, Cambridge, MA, U.S.A.
Ultrasound Imaging and Intervention Department, May 2019–September 2019.

2018–2019 **Product Development Engineer Intern** UbiRos, Natick, MA, U.S.A.

2018 **Research and Innovation Engineer II & III** Advansys ESC, Cairo, Egypt
Innovation Department, March 2018–August 2018.

2016–2017 **Mechanical Engineer I & II** Advansys ESC, Cairo, Egypt
Mechanical Department, September 2016–February 2017.

2015–2016 **Mechanical Engineer** Technical Office of Hydraulics (TO-Hydraulics), Cairo, Egypt
Maintenance Department.

HONORS AND AWARDS

2025 Best Paper and Oral Presentation Award, 14th Conference on New Technologies for Computer/Robot Assisted Surgery (CRAS).

2025 Runner-Up Best Poster Award, The Unconventional Robotics Symposium, TU Delft.

2024 Maurice-Godet Award for Best Oral Presentation and Paper in Tribology, 49th Leeds-Lyon Symposium on Tribology.

2024 Best Poster Award, BioMechanical Engineering Department Conference (BMechE).

2024 Best Poster Award, Dutch Soft Robotics Symposium (DSRS).

2024 RoboSoft24 Travel Grant for Excellence, IEEE Robotics and Automation Society (RAS).

2018–2020 WPI graduate student scholarship for academic excellence, Worcester Polytechnic Institute.

2018 Certificate of Appreciation for Outstanding Performance, Advansys ESC.

2016 MSc graduate scholarship for academic excellence, Faculty of Engineering, Cairo University.

2014 Distinguished Undergraduate Student Award, Cairo University and Engineering Company for Industrial Services.

GRANTS

2018–2020 Fulbright Foreign Student Program Grant, U.S. Department of State and Fulbright Commission in Egypt. Full scholarship to pursue MSc studies in the U.S.A.; selected among the top 1% of applicants in Egypt.

2016 Research support grant, Science and Technology Development Fund (STDF), Egypt. Collaborative research grant between Advansys ESC and Ain Shams University on robotic assistive rehabilitation.

PUBLICATIONS

Journal Publications and Manuscripts

J1. M. A. Atalla, A. S. Sekar, R. van Starkenburg, D. J. Jager, A. Sakes, M. Wiertlewski, and P. Breedveld, "OSCAR: An Ovipositor-Inspired Self-Propelling Capsule Robot for Colonoscopy," under review. Preprint: doi:10.48550/arXiv.2602.15309.

J2. M. A. Atalla, J. Nieuwenhuis, A. Martin, X. Wang, A. Canden, M. J. Carré, R. Lewis, A. Sakes, and M. Wiertlewski, "Active Lubrication of Transluminal Medical Instruments," under review. Preprint: doi:10.48550/arXiv.2506.07225.

J3. M. A. Atalla, J. J. Tuijp, M. Wiertlewski, and A. Sakes, "Toward Variable-Friction Catheters Using Ultrasonic Lubrication," *IEEE Transactions on Medical Robotics and Bionics*, vol. 6, no. 4, pp. 1375–1381, 2024. doi:10.1109/TMRB.2024.3464672.

J4. V. G. Kortman, Y. Verberne, M. A. Atalla, and A. Sakes, "Advancements in aspiration catheter tip design for thrombectomy: a comprehensive patent review," *Frontiers in Medical Technology*, vol. 6, article 1388638, 2024. doi:10.3389/fmedt.2024.1388638.

J5. A. Sabbadini, M. A. Atalla, and M. Wiertlewski, "Simple and Fast Locomotion of Vibrating Asymmetric Soft Robots," *Soft Robotics*, vol. 10, no. 6, pp. 1199–1208, 2023. doi:10.1089/soro.2022.0209.

J6. M. A. Atalla, R. A. J. van Ostayen, A. Sakes, and M. Wiertlewski, "Incompressible squeeze-film levitation," *Applied Physics Letters*, vol. 122, no. 24, p. 241601, 2023. doi:10.1063/5.0149501.

J7. N. E. Pacheco, J. B. Gafford, M. A. Atalla, R. J. Webster, and L. Fichera, "Beyond Constant Curvature: A New Mechanics Model for Unidirectional Notched-Tube Continuum Wrists," *Journal of Medical Robotics Research*, vol. 6, no. 1–2, 2021. doi:10.1142/S2424905X21400043.

J8. M. H. Mohamed, S. H. Wagdy, M. A. Atalla, A. R. Youssef, and S. A. Maged, "A proposed soft pneumatic actuator control based on angle estimation from data-driven model," *Proceedings of the Institution of Mechanical Engineers, Part H: Journal of Engineering in Medicine*, vol. 234, no. 6, pp. 612–625, 2020. doi:10.1177/0954411920911277.

Conference Publications

C1. M. A. Atalla, C. R. Derla, P. Canyelles Pericàs, and G. Krijnen, "Toward 3D Printable Non-Planar Electroadhesive Structures for Active Anchoring," in *Proceedings of IEEE SENSORS 2026*, Rotterdam, The Netherlands, 2026, (accepted). Preprint: doi:10.48550/arXiv.2609.18700.

C2. M. A. Atalla, J. Cumming, D. v. Bemmel, P. Breedveld, M. Wiertlewski and A. Sakes, "Grip as Needed, Glide on Demand: Ultrasonic Lubrication for Robotic Locomotion," *2026 IEEE International Conference on Robotics and Automation (ICRA)*, Vienna, Austria, 2026, pp. 23957-23964, doi:10.1109/ICRA57385.2026.11696075.

C3. M. A. Atalla, J. Nieuwenhuis, A. Martin, A. Canden, X. Wang, M. J. Carré, R. Lewis, M. Wiertlewski, and A. Sakes, "Ultrasonically-lubricated catheters: miniaturization and ex-vivo tissue validation," in *Hamlyn Symposium on Medical Robotics (HSMR) Proceedings*, 2024. doi:10.31256/HSMR2024.10.

C4. M. A. Atalla, F. Trauzettel, S. P. van Gelder, P. Breedveld, M. Wiertlewski, and A. Sakes, "Mechanically-Inflatable Bio-Inspired Locomotion for Robotic Pipeline Inspection," in *Proceedings of the 2024 IEEE 7th International Conference on Soft Robotics (RoboSoft)*, San Diego, CA, USA, 2024, pp. 586–592. doi:10.1109/RoboSoft60065.2024.10521968.

C5. M. A. Atalla, J. J. Tuijp, M. Wiertlewski, and A. Sakes, "Ultrasonically-Lubricated Catheters: A Proof of Concept," in *Hamlyn Symposium on Medical Robotics (HSMR) Proceedings*, London, UK, 2023. doi:10.31256/HSMR2023.12.

Conference Abstracts

A1. M. A. Atalla, A. Sekar, R. van Starkenburg, D. Jager, A. Sakes, and P. Breedveld, "OSCAR: An Ovipositor-Inspired Capsule Robot for Colonoscopy – A Proof of Concept," 14th Conference on New Technologies for Computer/Robot Assisted Surgery (CRAS), Lisbon, Portugal, 2025.

A2. M. A. Atalla, T. de Visser, A. Sakes, M. Wiertlewski, and Y. Vardar, "Electroadhesive Stabilization of Soft Robotic Catheters," IEEE 8th International Conference on Soft Robotics (RoboSoft), Lausanne, 2025.

A3. M. A. Atalla, J. Nieuwenhuis, A. Martin, A. Canden, X. Wang, M. J. Carré, R. Lewis, M. Wiertlewski, and A. Sakes, "Friction modulation of catheters using ultrasonic waves," 49th Leeds-Lyon Symposium on Tribology, 2024.

A4. M. A. Atalla, F. Trauzettel, S. P. van Gelder, P. Breedveld, M. Wiertlewski, and A. Sakes, "Mechanically-Inflatable Bio-Inspired Locomotion for Robotic Pipeline Inspection," Dutch Soft Robotics Symposium (DSRS), 2024.

A5. M. A. Atalla, R. A. J. van Ostayen, A. Sakes, and M. Wiertlewski, "On the Squeeze Film Levitation Phenomenon in Incompressible Liquid Environments," APS March Meeting, 2023.

A6. M. A. Atalla, A. Sakes, and M. Wiertlewski, "Near-Field Acoustic Levitation in Liquid Environment," 2022 IEEE International Ultrasonics Symposium (IUS), Venice, Italy, 2022.

Patents

P1. M. A. Atalla, J. Nieuwenhuis, M. Wiertlewski, and A. Sakes, *Medical device for intraluminal intervention*, Dutch patent NL2036368B1, granted, published Jun. 4, 2025.

P2. M. A. Atalla, R. Q. Erkamp, and M. M. Nguyen, *Mechanical Shear Wave Generation for Ultrasonic Elastography*, U.S. Patent US12364463B2, granted.

SCIENTIFIC PRESENTATIONS

Oral Presentations

- "Designing 'With' Friction: A New Paradigm Toward Programmable Physical Interaction," Embodied Intelligence Conference, 2026.
- "OSCAR: an Ovipositor-Inspired Capsule Robot for Colonoscopy: A Proof of Concept," CRAS, Lisbon, Portugal, 2025. Best Paper and Oral Presentation Award.
- "Active Lubrication of Endoluminal Instruments Using Ultrasonic Waves," Dutch Biomedical Engineering Conference, Egmond aan Zee, 2025.
- "Friction modulation of catheters using ultrasonic waves," 49th Leeds-Lyon Symposium on Tribology, 2024. Maurice-Godet Award for Best Oral Presentation and Paper in Tribology.
- "Ultrasonically-lubricated catheters: miniaturization and ex-vivo tissue validation," HSMR, 2024.
- "Active Lubrication of Endoluminal Instruments Using Ultrasonic Waves," Health Tech Open Day, TU Delft, 2024.
- "Ultrasonically-Lubricated Catheters: A Proof-of-Concept," HSMR, London, UK, 2023.
- "On the Squeeze Film Levitation Phenomenon in Incompressible Liquid Environments," APS March Meeting, 2023.

Poster Presentations

- "Unconventional Robotic Locomotion: The Ultrasonic Lubrication Way," The Unconventional Robotics Symposium, TU Delft, 2025. Runner-Up Best Poster Award.
- "Electroadhesive Stabilization of Soft Robotic Catheters," IEEE International Conference on Soft Robotics (RoboSoft), Lausanne, 2025.
- "Toward Variable-Friction Catheters Using Ultrasonic Lubrication," BioMechanical Engineering Department Conference (BMechE), 2024. Best Poster Award.
- "Mechanically-Inflatable Bio-Inspired Locomotion for Robotic Pipeline Inspection," Dutch Soft Robotics Symposium (DSRS), 2024. Best Poster Award.
- "Mechanically-Inflatable Bio-Inspired Locomotion for Robotic Pipeline Inspection," IEEE International Conference on Soft Robotics (RoboSoft), San Diego, CA, USA, 2024.
- "Near-Field Acoustic Levitation in Liquid Environment," IEEE International Ultrasonics Symposium (IUS), 2022.

Invited Talks

- "Safe and Functional by Design: Engineering Robots for Intelligent Physical Interaction," Robotics (RBT) Colloquium, Eindhoven University of Technology (TU/e), The Netherlands.
- "The Body Thinks Too: Physical Intelligence for Medical Robotics – in the Age of AI," ORSI Innotech Surgical AI & Telesurgery Days 2026, Ghent, Belgium.

SUPERVISION

Role: daily supervisor and mentor.

- Joan Castelló Belenguer, MSc thesis (2025), "Active Sealing using Squeeze Film Levitation," co-supervised with R. van Ostayen.
- Jack Cummings, MSc thesis (2024), "Ultrasonically-augmented inchworm locomotion," co-supervised with A. Sakes.
- Tim de Visser, MSc thesis (2024), "Stabilization of endoscopes using electroadhesion," co-supervised with Y. Vardar.
- Daan van Bemmel, MSc thesis (2024), "Efficient Locomotion for Sliding Robots," co-supervised with P. Breedveld.
- Jelte Nieuwenhuis, MSc thesis (2023), "Ultrasonic friction modulating rings for endoscopes," co-supervised with A. Sakes.
- Rick Matthijssen, MSc thesis (2023), "Multi-sample cardiac bioptome," co-supervised with A. Sakes.
- Anand Sekar, MSc thesis (2023), "Design of Intestine Inspection Device," co-supervised with P. Breedveld.
- Jeroen Tuijp, MSc thesis (2022), "Friction Modulation for Endoscopes: A Proof of Concept," co-supervised with M. Wiertlewski.

TEACHING

- Lecturer, AI-Image Guided Robotics (graduate course), Spring 2025. Delivered two lectures.
- Teaching assistant, Bio-Inspired Design (graduate course), instructor: P. Breedveld, Fall 2023.
- Grader, Fundamentals of Biomedical Engineering, instructor: A. Sakes, Fall 2023.
- Teaching assistant, Control in Human-Robot Interaction (graduate course), instructor: M. Wiertlewski, Spring 2021.

WORKSHOP ORGANIZATION

- M. A. Atalla, "From Research Idea to Funded Proposal: Grant Acquisition for (Unconventional) Early-Career Researchers," The Unconventional Robotics Symposium, TU Delft, The Netherlands, September 14, 2026. Role: organizer.
- M. A. Atalla, S. Sharma, T. Morimoto, T. Nanayakkara, R. J. Webster III, and A. Menciassi, "Soft $\neq$ Safe? Rethinking Softness in Medical Robotics," 2026 IEEE International Conference on Robotics and Automation (ICRA), Vienna, Austria, 2026. Role: lead organizer.
- N. N. Dei, A. B. Dawood, D. Osman, M. A. Atallaand K. Althoefer, "Soft Robotic Systems for Diagnostics and Therapy in Minimally Invasive and Endoluminal Interventions," Hamlyn Symposium on Medical Robotics (HSMR), London, 2026. Role: co-organizer.
- B. Calmé, L. Mocellin, M. A. Atalla, Q. Boehler, C. Bergeles, T. Horeman, and A. Menciassi, "Smart by Structure: Designing Surgical Devices for Equity and Sustainability," Hamlyn Symposium on Medical Robotics (HSMR), London, 2026. Role: co-organizer.
- M. A. Atalla, L. Lorenzon, M. Maselli, M. Singh, L. Paterno, and K. Suzumori, "Soft Robotics Mimicking Human Body for Medical Applications," 8th IEEE International Conference on Soft Robotics (RoboSoft), Lausanne, 2025. Role: lead co-organizer.
- M. A. Atalla, M. Finocchiaro, Y. Ge, L. Paterno, J. Kinross, M. Abayazid, and T. Nanayakkara, "From Research to Clinical Practice: Bridging the Gap in Robotic Colonoscopy," Hamlyn Symposium on Medical Robotics (HSMR), London, 2024. Role: lead co-organizer.
- M. A. Atalla, D. Wu, L. Fichera, and A. Sakes, "Bio-inspired medical robotics: from evolution to revolution in healthcare," Hamlyn Symposium on Medical Robotics (HSMR), London, 2024. Role: lead organizer.

COMMUNITY LEADERSHIP

Corresponding Founding Co-Chair IEEE RAS Working Group on Safety in Surgical Robotics
Hosted by the IEEE RAS Technical Committee on Surgical Robotics; coordinating an international forum on safety definitions, evidence, and evaluation in surgical robotics. Co-chairs: Di Wu, Julie Legrand, Linda Paternò, and Tim Horeman.

ACADEMIC COMMUNITY SERVICE

- Associate Editor, 2026 11th IEEE RAS/EMBS International Conference on Biomedical Robotics and Biomechatronics (BioRob 2026).
- Reviewer for IEEE Transactions on Robotics (TRO), IEEE Transactions on Medical Robotics and Bionics (TMRB), IEEE Robotics and Automation Letters (RA-L), IEEE International Conference on Robotics and Automation (ICRA), IEEE/ASME Transactions on Mechatronics (TMECH), Applied Physics Letters (APL), and Physics of Fluids (PoF).
- Volunteer, soft robotics workshop ("Make your own soft robot"), RoboCup 2024 Science Festival, Eindhoven, The Netherlands, July 17–21, 2024.
- Organizing committee member, World Haptics Symposium, 2023.
- Session chair, American Physical Society (APS) March Meeting, 2023.
- Organizing committee member, Dutch Soft Robotics Summer School and Symposium, 2022.
- Board Member, Fulbright Association Massachusetts Chapter, MA, U.S.A., 2020.
- Class Representative, Mechanical Engineering Department, Cairo University, Egypt, 2014.

RESEARCH OUTREACH AND MEDIA COVERAGE

- TU Delft news feature, "New catheter technology promises safer and more efficient treatment of blood vessels," November 2024.
- TU Delft research outreach feature, "Designing better catheters – with both more and less friction," June 2022.

REFERENCES

Paul Breedveld
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Robert J. Webster III
Richard A. Schroeder Professor of Mechanical Engineering, Vanderbilt University, U.S.A.
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